# OJAS PAL

ojaspal.dev@gmail.com | LinkedIn | GitHub

## PROFESSIONAL SUMMARY

AI & Data Science undergraduate building agentic workflows, RAG pipelines, and LLM apps on Google Cloud across data preprocessing, vector search, fine-tuning, and containerized serving for production-style GenAI systems.

## EDUCATION

**University School of Automation and Robotics (USAR), GGSIPU East Campus, Delhi** **2024 – 2028**
B.Tech in Artificial Intelligence & Data Science (AI-DS)

## TECHNICAL SKILLS

**Languages:** Python, C++

**AI & GenAI Frameworks:** LangChain, Gemini API, Groq API, RAG (Retrieval-Augmented Generation), Ollama

**Agentic AI:** LangGraph, Tool Calling, Structured Outputs, Human-in-the-Loop, Agent Orchestration

**ML & Deep Learning:** PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Pandas, NumPy, NLTK

**Concepts:** Computer Vision, NLP, Recommender Systems, Feature Engineering

**Databases & Cloud:** Pinecone, Cloud Firestore, Firebase Auth, Google Cloud Run, Secret Manager

**Backend & UI:** FastAPI, Flask, Next.js, REST APIs, Streamlit

**Developer Tools:** Docker, Git, GitHub, Jupyter Notebook, Google Cloud IAM

## PROJECTS

**GitResolve: Autonomous GitHub Issue Repair Agent** | GitHub **November 2026**
*Python, LangGraph, FastAPI, Next.js, Gemini API, Docker, Firestore, Cloud Run, Compute Engine*

- Built a 13-stage LangGraph agent with 2 human approval gates and a ModelRouter enforcing per-task budgets (50 model calls, 6 repair attempts) to localize faults, reproduce bugs, and generate minimal patches.
- Verified candidate patches in a hardened Docker sandbox (--network none, read-only rootfs, 300s timeout) with pytest and SHA-256 diff hash locking before opening draft GitHub PRs (30 unit tests passing).
- Deployed Next.js on Vercel, FastAPI on Cloud Run, and a GCE worker (4 vCPU, 16 GB RAM) via Firestore.

**ML-DL Portfolio: 11 Models in One Streamlit App** | GitHub **October 2026**
*Python, Streamlit, Scikit-Learn, CatBoost, PyTorch, TensorFlow, Keras, OpenCV*

- Rebuilt 11 notebooks (8 ML, 3 DL) into one Streamlit app with @st.cache_resource (.joblib, .pth, .keras).
- Replaced an ~8 GB similarity matrix (32,833 songs) with per-query cosine similarity to fit 2.7 GB RAM.
- Fine-tuned EfficientNet-B0 in two stages on 46 pet breeds merged from Oxford-IIIT Pet and Stanford Dogs, reaching 93.6% validation accuracy and 25–30 ms CPU inference on Streamlit Cloud.

**Medical RAG Chatbot** | GitHub **September 2026**
*Python, LangChain, Groq API, Pinecone, HuggingFace, Flask, Google Cloud Run*

- Split a 637-page medical encyclopedia into 5,800+ chunks, embedded them with bge-small-en-v1.5 (384-dim) and indexed them in Pinecone, returning top-2 lookups in under 45 ms.
- Passed retrieved chunks to gpt-oss-20b on Groq via LangChain with strict source-bound medical prompts.
- Deployed on Cloud Run (2 GiB, 2 vCPU, 8 threads) with lazy-loaded embeddings for sub-second startup.

**Personal Gemini Journal** | GitHub **September 2026**
*Gemini API, Firestore, Firebase Auth, Google Cloud Run, Secret Manager*

- Enforced Firebase Auth ID token + revocation checks on all 9 REST endpoints with /users/{uid} isolation.
- Cached Gemini keys from Secret Manager for 1 hour in RAM under a 3-role least-privilege Cloud Run account.
- Built "Ask My Journal" search over past entries using Gemini embeddings + keyword scoring, injecting the last 3 entries as context with zero extra API calls.

## CERTIFICATIONS

- Google Cloud Gen AI Academy, APAC Edition, Cohort 3 (Hack2Skill)
- IBM SkillsBuild AI Automation & Intelligent Solutions Internship